Table 8: Breakdown of AlpacaEval test set instructions by expected response length.

Expected Length	Number	Percentage
1-3 sentences	361	44.84%
1 paragraph	269	33.42%
1 sentence	143	17.76%
2 paragraphs	31	3.85%
3 or more paragraphs	1	0.13%

A.4 Preference optimization outperforms augmenting with positive examples only

We also tried an alternative self-training procedure of adding high-quality self-instruction creation examples to supervised fine-tuning (without preference optimization), rather than DPO. In this variant, we add additional examples of (instruction prompt, response) curated by the model to the seed set for supervised fine-tuning, following other approaches [Li et al., 2024, Adolphs et al., 2023, Gulcehre et al., 2023], rather than constructing preference data. In this setup we only add examples where the candidate response was evaluated to give a perfect score of $r_i^n = 5$. Unfortunately we could not find a configuration where this approach helped. For example, adding 11,254 such examples that scored 5 out of 5, and optimizing the mixing weight in training, still yielded a head to head with the SFT Baseline of 29% wins vs 30% wins, i.e., no improvement.

A.5 Augmented Prompt Generation Using Newly Trained Models

In our experiments, for time efficiency, we have created a fixed pool of augmented prompts in advance using ChatLlama 70B. In a real interactive system, ideally, those prompts could come from real users so that we can ensure the models are trained to align with real user requirements. Here, we also examine whether our newly trained Self-Rewarding models in each iteration can generate new prompts through in-context learning, instead of using ChatLlama 70B. To check this, we constructed 30 prompts with in-context examples using the original seed IFT data as described in Section 2.2 and tested whether M_1, M_2 and M_3 still possess in-context learning ability and can generate high quality instructions. According to manual inspection, all models can generate novel instructions given in-context examples in all 30 cases. However, for $M2$ and $M3$, the model is likely to first generate a few instructions, then generate a separator, and then start responding to the instructions, so some postprocessing might be necessary.

A.6 AlpacaEval Test Sample Clustering

We used the GPT-4 (`gpt-4-1106-preview`) model to categorize the instructions in the AlpacaEval test set into clusters from three perspectives: (1) instruction category, (2) instruction complexity, and (3) expected response length. To obtain instruction categories for the AlpaceEval test set, we used the prompt in Figure 9 and obtained 20 categories in total. Then, to cluster the instructions into different groups, we use the prompt in Figure 10 for each test example. The corresponding statistics are given in Table 6, Table 7, Table 8. The fine-grained results on instruction complexity and expected response length are given in Figure 11.

Table 9: **NLP Benchmarks**. Self-Rewarding models mostly tend to maintain performance compared to the Llama 2 base model and the SFT Baseline, despite being fine-tuned on very different instruction-following prompts.

	Commonsense Reasoning					**Math**	**World Knowledge**		
	ARC easy	ARC challenge	HellaSwag	SIQA	PIQA	GSM8K (em)	MMLU (macro_avg/acc)	OBQA (acc_comp)	NQ (em)
Llama 2	80.20	57.40	85.30	50.70	82.80	56.80	68.90	60.20	25.30
SFT Baseline	76.49	55.97	85.17	51.48	82.59	50.72	69.76	57.80	34.35
M_1	78.14	57.51	84.99	53.02	82.92	60.27	69.34	57.60	35.48
M_2	74.84	54.51	84.27	51.23	81.94	59.29	69.31	57.60	33.07
M_3	72.35	53.13	83.29	49.28	80.79	57.70	69.37	58.40	31.86

Table 10: **MT-Bench Fine-grained Results**. We list our models' performance on each problem category. Self-reward is especially effective in improving the model's ability in writing, role-playing, extraction, and STEM tasks.

	Writing	**Roleplay**	**Reasoning**	**Math**	**Coding**	**Extraction**	**STEM**	**Humanities**	**Overall**
SFT	8.83	8.15	5.30	3.00	3.50	6.90	9.18	9.95	6.85
M1	9.10	7.65	4.35	3.05	4.10	7.20	8.93	9.85	6.78
M2	9.10	8.00	4.60	3.30	4.25	7.65	9.40	9.80	7.01
M3	9.58	8.73	4.80	3.50	4.20	7.80	9.45	9.95	7.25

A.7 NLP Benchmark Results and MT-Bench Results

We provide the detailed model performance on a number of NLP benchmarks in Table 9 and on MT-Bench in Table 10. In particular, some NLP benchmarks including ARC-Challenge, HellaSwag, SIQA, PIQA, and OBQA are all text completion tasks. In these tasks, given the multiple choice options, we choose the option corresponding to the highest log probability scored by the models as the final answer. As such, the objective of these particular tasks is quite different from what our algorithm tries to optimize, so the results on these tasks may not reflect the true capability of our models.